CHAPTER 1

Sequences and proof by induction

S2, S4, S5, S7, S10, S11, S13

Before you begin · Sequences and index notation

Without a calculator, in six minutes. Answer briefly. Use the "reference sequences" sheet and E1-E2. Then choose the targeted review shown under the questions.

1. Calculate u_1 then u_2 if $u_0 = 3$ and $u_{n+1} = 2u_n - 1$.

2. Distinguish u_{n+1} and $u_n + 1$.

3. Give 1 $+ 2 + \cdots + 10$.

4. Write $\sum_{k=0}^{3} 2^k$.

5. Determine the multiplier for a decrease of 20%.

6. Give the first three values of `n` in `for n in range(3)` .

Answers and remediation

5, 9; these are two different expressions; 55; 1 $+ 2 + 4 + 8$; 0,8 ; 0, 1, 2. Have students verbalise index and value before any induction.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Substitute the previous value step by step; R01. **2.** Write an index as a subscript and an addition on the line; R01. **3.** Pair the first and last terms; R09 to compare two sums. **4.** Expand the four terms; R08 checks how many there are. **5.** Write 100 minus 20 over 100; R03. **6.** List the values of range before the calculation; R07. The R questions serve as review prompts with the teacher, even if their whole-class routine is planned later.

Objectives

Calculate terms and a sum; write an induction proof; establish monotonicity and bounds; use an auxiliary sequence.

Course notes

Define a sequence and calculate

A real sequence is a function defined on $\mathbb{N}$, or on the integers from some index on n_0. We write u_n for its term of index n. An explicit formula gives u_n directly; a recurrence relation requires also an initial term. For example, $u_0 = 2$ and $u_{n+1} = 3u_n - 1$ give $u_1 = 5$, $u_2 = 14$.

Reference sequences

If $u_{n+1} = u_n + r$, then $u_n = u_p + (n-p)r$. If $u_{n+1} = qu_n$, then $u_n = u_p q^{n-p}$ for $n \geq p$.

$$\sum_{k=p}^{n} u_k = (n-p+1)\frac{u_p + u_n}{2} \quad \text{(arithmetic)}, \qquad \sum_{k=p}^{n} u_k = u_p \frac{1-q^{n-p+1}}{1-q} \quad (q \neq 1).$$

When $q = 1$, the second sum is $(n-p+1)u_p$.

Proof by induction

To prove $P(n)$ for every integer $n \geq n_0$: check $P(n_0)$; fix an arbitrary integer $n \geq n_0$ and assume $P(n)$ true; deduce $P(n+1)$; conclude by the principle of induction. The hypothesis is not the conclusion: it can only be used at the fixed index.

Fully written-up example. Let $u_0 = 1$, $u_{n+1} = 2u_n + 3$. For $n \in \mathbb{N}$, let $P(n)$: "$u_n = 4 \times 2^n - 3$" .

Base case. $4 \times 2^0 - 3 = 1 = u_0$; $P(0)$ is true.

Inductive step. Fix $n \in \mathbb{N}$ and suppose $P(n)$ true. By the definition of the sequence, then by this hypothesis,

$$\begin{aligned} u_{n+1} &= 2u_n + 3 \\ &= 2(4 \times 2^n - 3) + 3 \\ &= 8 \times 2^n - 3 = 4 \times 2^{n+1} - 3. \end{aligned}$$

This is $P(n+1)$: the property is hereditary.

Conclusion. By the principle of induction, $u_n = 4 \times 2^n - 3$ for every $n \in \mathbb{N}$.

Proof: sum of integers. Let $P(n)$: "$\sum_{k=0}^{n} k = \frac{n(n+1)}{2}$ ". For $n = 0$, both sides equal 0: $P(0)$ is true. Fix $n \geq 0$ and suppose $P(n)$ true. Then

$$\begin{aligned} \sum_{k=0}^{n+1} k &= \sum_{k=0}^{n} k + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{(n+1)(n+2)}{2}. \end{aligned}$$

We proved $P(n+1)$. By induction, the formula holds for every $n \in \mathbb{N}$.

Overall behaviour

The sequence is increasing if $u_{n+1} \geq u_n$ for every index, decreasing if the inequality is reversed. It is bounded above if there is a real number M such that $u_n \leq M$ for all n ; bounded below: some m such that $u_n \geq m$; bounded if it is both bounded below and bounded above. To study monotonicity, we can study the sign of $u_{n+1} - u_n$. If all the terms are strictly positive, we can compare $\frac{u_{n+1}}{u_n}$ to 1. The study of a function f such that $u_n = f(n)$ is another possibility.

Arithmetico-geometric sequence

For $u_{n+1} = au_n + b$, $a \neq 1$, solve $\ell = a\ell + b$, then set $v_n = u_n - \ell$. We get $v_{n+1} = av_n$, so

$$u_n = \ell + (u_0 - \ell)a^n, \qquad \ell = \frac{b}{1-a}.$$

If $a = 1$, the sequence is arithmetic: do not divide by $1 - a$.

Example. If $u_0 = 10$ and $u_{n+1} = 0{,}8u_n + 6$, then $\ell = 30$ and $u_n = 30 - 20(0{,}8)^n$. We have $10 \leq u_n < 30$ and $u_{n+1} - u_n = 4(0{,}8)^n > 0$.

Pitfall. Checking a property on a hundred terms does not prove it. An increasing sequence is not necessarily positive; a bounded sequence is not necessarily monotonic.

Understand, try, revisit

1.1 Index, value and update

First use: S2 and S4.

Definition · A sequence is not a finite list

The index n denotes an integer; u_n is a real number. The representative point is $(n; u_n)$. A line connecting the points can help with reading but does not define terms at non-integer indices.

Worked example · Two expressions that look alike

For $u_0 = 2$ and $u_{n+1} = 3u_n - 1$, we get $u_1 = 5$, $u_2 = 14$, $u_3 = 41$. Thus $u_2 + 1 = 15$, but $u_{2+1} = 41$. In a loop, the variable `u` first contains u_n; the instruction `u = 3*u-1` replaces it with u_{n+1}.

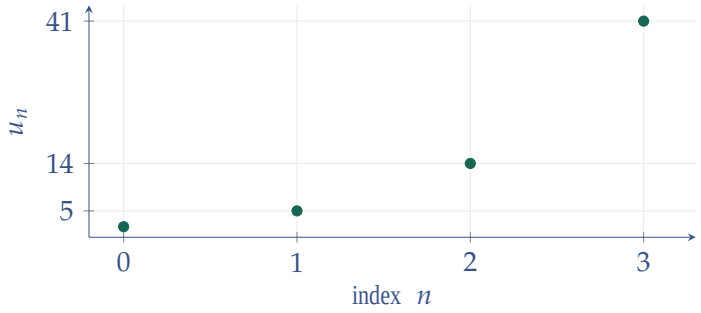


Figure 1.1. Read the term of index 2: its y-coordinate is 14, not 2.

A1 · Immediate application For $v_0 = 1$, $v_{n+1} = 2v_n + 1$, build a table of indices 0 to 3, then place the four points. Calculate separately $v_2 + 1$ and v_{2+1}.

A2 · Partial guidance A loop must calculate $u_0 + u_1 + u_2 + u_3$ for the sequence in the example. It starts at `u=2; total=2`. How many updates are needed? Write the order of the instructions and explain why adding before the update would count twice u_0.

1.2 A local hypothesis, a universal conclusion

First use: S5 and S7.

Worked example · Induction for an inequality

Let $w_0 = 1$ and $w_{n+1} = \frac{1}{2}w_n + 2$. Let us prove $P(n) : 1 \leq w_n < 4$. Initially $w_0 = 1 \in [1; 4[$. Fix $n \in \mathbb{N}$ and suppose $P(n)$. Since $\frac{1}{2} > 0$, $\frac{1}{2} \leq \frac{1}{2}w_n < 2$, so $\frac{5}{2} \leq w_{n+1} < 4$. Since $\frac{5}{2} \geq 1$, this implies $P(n+1)$. The principle of induction gives the bounds at every index.

A3 · Proof to be completed For $a_0 = 2$, $a_{n+1} = 2a_n + 1$, complete: " $P(n)$ means $a_n = 3 \times 2^n - 1$. At index 0, . . . Fix . . . and assume . . . Then $a_{n+1} = \ldots = 3 \times 2^{n+1} - 1$. Hence . . ."

A4 · Fix a proof A student writes: "If $u_n \geq 2$, then $u_{n+1} = 2u_n \geq 4 \geq 2$. So all the terms are at least 2." He did not give u_0. Construct a counterexample, then state which data would complete the proof.

1.3 Count before summing

First use: S10.

Property · Number of terms

For integers $p \leq n$, the sum $\sum_{k=p}^{n} u_k$ contains $n - p + 1$ terms. This can be checked by writing out the first few and the last, before any formula.

Worked example · A shifted geometric sum

In $\sum_{k=3}^{7} 2^k$, there are five terms and the first equals 8. The sum is $8 \frac{1-2^5}{1-2} = 248$. For prove the formula without memorising it in isolation, write $S = 1 + q + \cdots + q^n$, then subtract $qS = q + \cdots + q^{n+1}$: $(1-q)S = 1 - q^{n+1}$. Dividing assumes $q \neq 1$.

A5 · Independent proof Prove by induction, for every $n \in \mathbb{N}$, $\sum_{k=0}^{n}(2k+1) = (n+1)^2$. Explain why the term added in the inductive step is $2 n + 3$, and not $2n + 1$.

1.4 Recognise a structure without confusing it with a limit

First use: S11 and S13.

Worked example · Choosing the shift

For $w_{n+1} = \frac{1}{2}w_n + 2$, the fixed value is 4. Subtracting 4 from both sides, $w_{n+1} - 4 = \frac{1}{2}(w_n - 4)$. Thus $w_n = 4 - 3\,2^{-n}$. The shift is not arbitrary: it removes the constant term. This explicit calculation does not yet assume a convergence theorem.

A6 · Choice of strategy A subscription costs 40 euros in the first year, then its cost increases each year by 5 euros. Another costs 40 euros and then increases by 10 % per year. Write both sequences, compare the first three years and calculate the cumulative costs over ten years. Justify the number of terms.

A first step: First name the type of the two changes. The tenth year corresponds to index 9 if the first is index 0.

One minute, spoken aloud. Explain the difference between "for every n, $u_n \leq 4$" and "there exists n such that $u_n \leq 4$". Give a sequence satisfying the second but not the first.

Delayed review · S20 then S41. Revisit $w_n = 4 - 3\,2^{-n}$ to prove its convergence; in probability, look again for the shift when a probability satisfies an affine relation.

Investigation

A tank topped up every day - 35 min.

A tank initially contains 100 litres of water. Each day, 20 % of the volume present is removed, then 30 litres are added. For every natural number n, let u_n the volume of water, in litres, after these two operations performed n times. Thus, $u_0 = 100$.

1. Calculate u_1 and u_2, then express u_{n+1} as a function of u_n.
2. Which value of the volume would remain unchanged after one day? Prove by induction that, for all $n \in \mathbb{N}$, $100 \leq u_n < 150$.

3. For every $n \in \mathbb{N}$, set $v_n = u_n - 150$. Show that (v_n) is geometric. Deduce the expressions for v_n and of u_n as a function of n.

4. Write a Python function with no argument that returns the first integer n for which $u_n > 145$. Run the program and interpret the result.

5. A student calculated the first hundred terms and notices that they are less than 150. Is this observation enough to prove that all the terms are? Justify.

Progressive exercises

E1 · Direct application. Consider the sequence (u_n) defined, for every natural number n, by $u_n = 5 - 3n$. Calculate u_4, then the sum $\sum_{k=0}^{20} u_k$.

E2 · Application and justification. Let (v_n) be defined by $v_0 = 3$ and, for every natural number n, $v_{n+1} = 2v_n$. Express v_n as a function of n, then calculate $\sum_{k=0}^{n} v_k$.

E3 · Method to organise. Prove by induction that, for every natural number n, $3^n \geq 2n + 1$.

E4 · Choice of strategy. Consider the sequence (u_n) defined on $\mathbb{N}$ by $u_n = \frac{n+1}{n+2}$. Prove that it is increasing and give bounds valid for all its terms.

E5 · Synthesis or proof. Prove by induction that, for every natural number n, $8^n - 1$ is divisible by 7. Congruences will not be used.

Work from the textbook or a past paper

Barbazo, p. 28, Exercises 22 and 25. In class, in S5: exercise 22, to express properties of sequences with a quantifier. At home after S11: exercise 25 in full, to link induction, upper bound and monotonicity. Write the base case, inductive step and conclusion separately.

Homework

DM1 - A sum and a threshold - 45 min, after S11.

Consider the sequence (u_n) defined by $u_0 = 2$ and, for every $n \in \mathbb{N}$, $u_{n+1} = \frac{1}{2}u_n + 3$.

1. Prove by induction that $2 \leq u_n < 6$ for all $n \in \mathbb{N}$. Deduce the direction of variation of (u_n).

2. Let $v_n = u_n - 6$ for all $n \in \mathbb{N}$. Show that (v_n) is geometric, then express u_n in function of n.

3. For every $n \in \mathbb{N}$, let $S_n = \sum_{k=0}^{n} u_k$. Express S_n as a function of n.

4. Using the explicit formula and powers of 2, determine the smallest integer n such that $u_n > 5.99$. Check that the previous integer does not work.

5. Express $6 - \frac{S_n}{n+1}$ and justify that this difference is strictly positive. Interpret the result in considering the mean of the first $n + 1$ terms; no limit calculation is required.

MCQ A · consolidation (after S7)

After S7: first induction proofs.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. If $u_0 = 2$ and $u_{n+1} = u_n + 3$, then u_2 is:

A. 5 B. 8

C. 11 D. 2^4

2. To prove $P(n)$ by induction from 0, the inductive step consists of:

A. assume $P(n+1)$ B. only check $P(0)$

C. assume $P(n)$ then prove $P(n+1)$ D. solve $P(n) = 0$

3. The sum $\sum_{k=0}^{4} 2^k$ includes:

A. 4 terms B. 5 terms

C. 8 terms D. 2^4 terms

4. If $u_{n+1} - u_n > 0$ for every n, then:

A. u is bounded above B. u is increasing

C. u converges D. $u_n > 0$

5. In `while u <= 10: u = u + 1`, the loop stops when:

A. $u = 10$ B. $u < 10$

C. $u > 10$ D. $u < 0$

MCQ B · review and choice of method (after S20)

After S20: monotone convergence and auxiliary sequence.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. A sequence is increasing and bounded above by 7. We can conclude that it:

A. tends to 7 B. converges

C. is positive D. is geometric

7. For $u_{n+1} = 0{,}5u_n + 4$, any equilibrium value satisfies:

A. $L = 0{,}5L + 4$ B. $L = u_0$

C. $L = 0{,}5 + 4L$ D. $L = 4^n$

8. If $v_n = u_n - 8$ and $u_{n+1} = 0{,}5u_n + 4$, then:

A. $v_{n+1} = v_n + 4$ B. $v_{n+1} = 0{,}5v_n$

C. $v_{n+1} = 8v_n$ D. $v_{n+1} = u_n$

9. "u_n belongs to $]5{,}99; 6{,}01[$" is equivalent to:

A. $u_n < 6$ B. $|u_n - 6| < 0{,}01$

C. $|u_n| < 6$ D. $u_n > 0{,}01$

10. A simulation of 20 terms proves that:

A. the statement is true for every n B. the sequence converges

C. the 20 calculated terms have the observed property D. induction is unnecessary

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Every increasing sequence converges.

2. If $u_n = 6 - 2^{-n}$, then $u_n < 6$ for all n.

3. Solve $L = f(L)$ is enough to prove that every sequence $u_{n+1} = f(u_n)$ converges.

Review of prior work

I can compute a term, a sum, set up an induction proof and study monotonicity using $u_{n+1} - u_n$.
To watch out for: confusing an index with a value; turning a numerical observation into a proof.

MCQ solutions

1. **B.** $u_1 = 5$ then $u_2 = 8$. Answers 5 and 11 confuse the indices; $2^4 = 16$ does not correspond to the additive relation.

2. **C.** We fix an index n and we assume $P(n)$ to prove $P(n+1)$. Checking the first index is the base case, and directly assuming the conclusion would be circular.

3. **B.** The indices are 0, 1, 2, 3, 4: five terms. The number of terms is last index minus first index plus one.

4. **B.** The sign of the differences proves the sequence is strictly increasing. $u_n = n - 100$ shows that neither positivity nor an upper bound nor convergence is implied.

5. **C.** The exit occurs when the condition $u \leq 10$ becomes false, so for $u > 10$. Reaching exactly 10 is not enough to stop the loop.

6. **B.** Monotone convergence gives a limit at most equal to 7. The constant sequence -1 refutes the limit 7 and positivity; $7 - 1/(n+1)$ is not geometric.

7. **A.** A value left unchanged by the update satisfies $L = 0{,}5L + 4$, i.e. $L = 8$. This equation alone does not prove convergence.

8. **B.** $v_{n+1} = 0{,}5u_n + 4 - 8 = 0{,}5(u_n - 8) = 0{,}5v_n$. Subtracting the fixed point removes the constant term.

9. **B.** The gap $|u_n - 6|$ measures the distance from u_n to 6. The strict inequality $|u_n - 6| < 0{,}01$ describes exactly the required open interval.

10. **C.** A finite calculation tells us only about the terms actually computed. A property for every index requires a proof; convergence requires a study of the whole sequence.

True/false. False: $u_n = n$ diverges. True: $2^{-n} > 0$, hence $6 - 2^{-n} < 6$. False: with $f(x) = -x$, the fixed point is 0 but $u_0 = 1$ generates $(-1)^n$, which diverges.

Solutions to the applications and guidance

Application 1. $v_0 = 1$, $v_1 = 3$, $v_2 = 7$, $v_3 = 15$. The points are $(0;1), (1;3), (2;7), (3;15)$. $v_2 + 1 = 8$ but $v_{2+1} = 15$.

Application 2. Three updates: repeat `u=3*u-1`, then total=total+u . We get $2 + 5 + 14 + 41 = 62$. The reverse order would successively add 2,5,14 to the total already initialised to 2 and would omit 41.

Application 3. At index 0, $3 \times 2^0 - 1 = 2 = a_0$. Fix $n \in \mathbb{N}$ and suppose $a_n = 3 \times 2^n - 1$. Then $a_{n+1} = 2(3 \times 2^n - 1) + 1 = 3 \times 2^{n+1} - 1$. The base case and the inductive step allow us to conclude for every $n \in \mathbb{N}$.

Application 4. $u_0 = 0$ gives the zero sequence: the inductive step is correct but never starts. Add $u_0 \geq 2$, then check $P(0)$. Do not confuse the validity of an implication with that of its hypothesis.

Application 5. At index 0: $1 = 1^2$. Under the hypothesis at index n, the sum at index $n+1$ equals $(n+1)^2 + 2(n+1) + 1 = (n+2)^2$. Conclude by induction. The new term has index $n+1$.

Application 6. Arithmetic sequence $a_n = 40 + 5n$; geometric sequence $b_n = 40(1{,}1)^n$. First costs : 40,45,50 and 40,44,48.4. Cumulative totals: $10(40 + 85)/2 = 625$ euros; $40 \frac{1{,}1^{10}-1}{0{,}1} \simeq 637{,}50$ euros. The three first years are not enough to predict the order of the cumulative costs.

Class discussion and differentiation

Have students compare their working in pairs before showing the formula. For the gap-fill proof, accept a personal wording provided the integer is fixed and the hypothesis is identifiable.
Then remove the prompts. At S13, go back to the base-case error with $u_0 = -1$. Observe the write-up and not only the numerical result.

Teaching notes and solutions

Guidance for the teacher

In S2, distinguish index and value of a term from the start. In S4, have students run a loop by hand before Python. In S5 and S7, have students say aloud "assume at index n, we prove at index $n+1$". Sums and divisibility come in S10, after DS1: do not assess them in the first written test. In S11, finding the fixed point is an algebraic technique; the theorem of convergence will be established in the next chapter. For struggling students, provide the property $P(n)$ and let them write out the three steps; for the more advanced, remove the hint about the auxiliary sequence.

Solutions to the activity

1. $u_0 = 100$ L. After one day, there are $0{,}8u_0$ L left, then 30 L are added: $u_1 = 110$ L. Similarly, $u_2 = 0{,}8 \times 110 + 30 = 118$ L. For every $n \geq 0$, $u_{n+1} = 0{,}8u_n + 30$.

2. The equilibrium volume L should satisfy $L = 0{,}8L + 30$, i.e. $L = 150$ L. Let us prove $P(n)$: "$100 \leq u_n < 150$". At index 0, $u_0 = 100 \in [100, 150[$. Fix $n \geq 0$ and suppose $P(n)$ true. Since $0{,}8 > 0$,
$$110 = 0{,}8 \times 100 + 30 \leq 0{,}8u_n + 30 < 0{,}8 \times 150 + 30 = 150.$$
Hence $100 \leq u_{n+1} < 150$ and $P(n+1)$ is true. By induction, the bounds hold for all $n \in \mathbb{N}$.

3. $v_{n+1} = u_{n+1} - 150 = 0{,}8u_n - 120 = 0{,}8(u_n - 150) = 0{,}8v_n$. The sequence (v_n) is geometric, with common ratio 0.8 and first term $v_0 = -50$. Thus $v_n = -50(0{,}8)^n$, then $u_n = 150 - 50(0{,}8)^n$.

4.
```
def premier_jour():
    n, u = 0, 100
    while u <= 145:
        u = 0.8*u+30
        n += 1
    return n
```

The loop returns 11. Indeed, $u_{10} \simeq 144{,}63 < 145$ and $u_{11} \simeq 145{,}71 > 145$. Moreover, $u_{n+1} - u_n = 10(0{,}8)^n > 0$: the threshold was not crossed earlier. Computing these two values is enough here to guarantee the loop stops, without yet requiring a convergence theorem.

5. The simulation only computes a finite number of terms. The inductive step, together with the base case, establishes the bounds for all integers; it is this that rules out exceeding 150 L.

E1. $u_4 = 5 - 3 \times 4 = -7$. The sequence is arithmetic with common difference -3 and $u_{20} = -55$. The sum has 21 terms: $\sum_{k=0}^{20} u_k = \frac{21(u_0+u_{20})}{2} = \frac{21(5-55)}{2} = -525$.

E2. The sequence is geometric with common ratio 2 and first term $v_0 = 3$. Hence $v_n = 3 \times 2^n$. The sum contains $n+1$ terms and equals $\frac{3(1-2^{n+1})}{1-2} = 3(2^{n+1} - 1)$.

E3. Let $P(n)$: "$3^n \geq 2n+1$". At index 0, $3^0 = 1 = 2 \times 0 + 1$. Fix $n \geq 0$ and suppose $P(n)$ true. Multiplying by $3 > 0$, $3^{n+1} \geq 6n + 3$. Now $6n + 3 - (2n+3) = 4n \geq 0$. Thus $3^{n+1} \geq 2(n+1) + 1$, which is $P(n+1)$. By induction, $P(n)$ is true for all $n \in \mathbb{N}$.

E4. For $n \geq 0$, $u_n = 1 - \frac{1}{n+2}$. We have $0 < \frac{1}{n+2} \leq \frac{1}{2}$, therefore $\frac{1}{2} \leq u_n < 1$. Moreover,
$$u_{n+1} - u_n = \frac{1}{n+2} - \frac{1}{n+3} = \frac{1}{(n+2)(n+3)} > 0.$$
The sequence is strictly increasing, bounded below by $\frac{1}{2}$ and bounded above by 1.

E5. Let $P(n)$: "there exists an integer k such that $8^n - 1 = 7k$ ". For $n = 0$, $8^0 - 1 = 0 = 7 \times 0 : P(0)$ is true. Fix $n \geq 0$ and suppose $P(n)$ true. So there exists an integer k such that $8^n - 1 = 7k$. Then

$$8^{n+1} - 1 = 8(8^n - 1) + 7 = 8 \times 7k + 7 = 7(8k + 1).$$

Since $8k + 1$ is an integer, $P(n+1)$ is true. By induction, 7 divides $8^n - 1$ for every $n \geq 0$.

Solutions to homework DM1

1. Let $P(n)$: "$2 \leq u_n < 6$". At index 0, $u_0 = 2$ satisfies the bounds. Fix $n \geq 0$ and suppose $P(n)$ true. Then $4 \leq \frac{u_n}{2} + 3 < 6$; in particular $2 \leq u_{n+1} < 6$. By induction, $P(n)$ is true at every index. We deduce $u_{n+1} - u_n = 3 - \frac{u_n}{2} = \frac{6-u_n}{2} > 0$. The sequence is strictly increasing.

2. The equation $L = \frac{L}{2} + 3$ gives $L = 6$. Let $v_n = u_n - 6$. Then $v_{n+1} = \frac{u_n}{2} - 3 = \frac{v_n}{2}$ & $v_0 = -4$. Hence $v_n = -\frac{4}{2^n}$ and $u_n = 6 - \frac{4}{2^n}$.

3. By linearity of sums and the geometric sum formula,

$$\begin{aligned} S_n &= 6(n+1) - 4\sum_{k=0}^{n}(\frac{1}{2})^k \\ &= 6(n+1) - 8(1 - 2^{-(n+1)}). \end{aligned}$$

4. $u_n > 5{,}99$ is equivalent to $\frac{4}{2^n} < 0{,}01$, i.e. $2^n > 400$. Since $2^8 = 256 \leq 400 < 512 = 2^9$ and (2^n) is increasing, the smallest index is 9.

5. The mean is less than 6 and its distance from 6 is

$$6 - \frac{S_n}{n+1} = \frac{8(1 - 2^{-(n+1)})}{n+1}, \qquad 0 < 6 - \frac{S_n}{n+1} < \frac{8}{n+1}.$$

To make this difference less than a precision $\varepsilon > 0$, it suffices to take $n + 1 > \frac{8}{\varepsilon}$. This is a quantitative control of the gap, which does not yet require a theorem on limits.

Solutions to the textbook or past-paper work

Exercise 22. The four translations are, for all $n \in \mathbb{N}$: $u_{n+1} \geq u_n$; $v_n \geq 5$; $w_n \leq -1$; $t_n > 0$. The monotonicity compares two consecutive terms; a bound compares each term with the same number.

Exercise 25. We have $u_0 = -5$ and $u_{n+1} = \frac{2}{3}u_n - 1$.

Base case. $u_0 = -5 \leq -3$: the property is true at index 0.

Inductive step. Fix $n \in \mathbb{N}$ and suppose $u_n \leq -3$. Multiplication by $\frac{2}{3} > 0$ preserves order, so $\frac{2}{3}u_n \leq -2$. Thus $u_{n+1} = \frac{2}{3}u_n - 1 \leq -3$.

Conclusion. By induction, $u_n \leq -3$ for every $n \in \mathbb{N}$. Then, $u_{n+1} - u_n = \frac{2}{3}u_n - 1 - u_n = -\frac{1}{3}u_n - 1$. The previous bound gives $-\frac{1}{3}u_n \geq 1$, hence $u_{n+1} - u_n \geq 0$. The sequence is increasing. The change of direction of the inequality here comes from the negative factor $-\frac{1}{3}$.